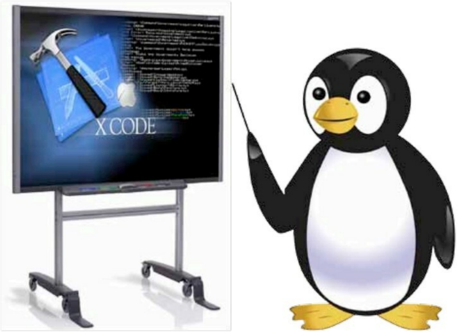

OS X Command Line Tools

Mac users may know of Xcode, the IDE (Integrated Development Environment) by Apple, which is used to develop software for OS X and iOS. One of its main components is Command-Line Tools, which this article covers.

Xcode was first seen in 2003; the latest stable release is v4.4, which is available free for Mac OS X Lion and OS X Mountain Lion in the Mac App Store. Xcode includes a modified version of the GNU Compiler Collection. Apple recently introduced the Command-Line Tools (CLT) as open source; these include the Apple LLVM compiler, linker and Make. The CLT can be used with or without Xcode. Along with these, Xcode also supports dozens of other programming languages, which include C, C++, Objective-C, Objective-C++, AppleScript, Java, Python, Ruby, Ada, Cocoa and Carbon. Apple claims to be the first major computer company to make open source a key part of its software strategy, and so has continued to use and release open source software.

Command line tools

Let's take a look at the major CLT utilities.

LLVM compiler

LLVM (Low-Level Virtual Machine) is a huge collection of tools and libraries that make it easy to build new compilers, and compile-time, link-time, run-time and 'idle-time' optimisers; Just-In-Time compilers (similar to Java); and a large, powerful and distinctive variety of compiler category programs. Some key features of the LLVM compiler include an extremely simple design, source-language independence, automated compiler debugging support, extensibility, stability and reliability. Apart from this, LLVM is also capable of generating relocatable machine code at any time of the program's life cycle. Initially written to replace the existing code generator in the GNU Compiler Collection (GCC) stack, it now supports various programming languages, including Ada, C, C++, Fortran, the use of many front-ends—and has had many GCC front-ends modified to work with it.

Linkers

A linker combines libraries and compiler-generated objects into an executable program. On UNIX-like systems, we use the term *loader* in place of linker. The linker is capable of including only the required objects from the library collection, when the library has diverse purposes. Hence, the linker plays a vital role in the CLT.